Knut Schmidt-Nielsen



Knut Schmidt-Nielsen grew up in the Norwegian town of Trondheim. His interest in the way animal physiology related to habitat was inherited from his grandfather who, years before Knut's birth, had released thousands of flounder (a marine fish) hatchlings into a freshwater lake. Although the fish thrived, they were unable to breed because their reproductive physiology was adapted for life in salt water.

Schmidt-Nielsen joined Duke University, North Carolina, in 1954. He built a climate-controlled space for keeping desert animals, where he considered the anatomy of camels, gerbils, and other species able to live for long periods without water. He also investigated the respiratory systems of birds and the buoyancy of fish. His 1960 textbook *Animal Physiology* is still a classic work.

Key works

1960 *Animal Physiology*

1964 *Desert Animals*

1972 *How Animals Work*

1984 *Scaling*

1998 *The Camel's Nose: Memoirs of a Curious Scientist*

See also: Evolution by natural selection • Ecological niches • Competitive exclusion principle • Ecological stoichiometry